

A ratchet that has already been tested: hypoxia, fibrosis and the chronicity of hard flaccid syndrome

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Tenth in a series; see companion papers on the neural, effector, sensory and structural conjectures.

Abstract

Background. Hard flaccid syndrome (HFS) rarely remits, and loss of morning and nocturnal erections is commonly reported. The conjecture examined here proposes a self-reinforcing loop: sustained contraction reduces perfusion, loss of nocturnal erections removes the tissue's principal reoxygenation, hypoxia raises transforming growth factor $\beta 1$, collagen accumulates, compliance falls, erections worsen, and nocturnal erections fall further — converting a functional disorder into a structural one, largely in one direction.

The pathway is real; the chain is long. Cavernous neurotomy in the rat increases hypoxia-inducible factor 1α , transforming growth factor $\beta 1$ and collagen I and III; oxygen-dependent prostanoid synthesis suppresses TGF- $\beta 1$ -induced collagen. But the ratchet requires six sequential links to hold, and its critics within the field it borrows from have noted that the effects of the oxygen tensions actually associated with a flaccid penis remain theoretical, with no study demonstrating in vivo derangement of endothelial or smooth muscle metabolism from a prolonged flaccid state.

This ratchet has already been tested in humans, and the result is not cited. Penile rehabilitation after radical prostatectomy is precisely an attempt to break it. Across randomised trials — REACTT, REINVENT and others — daily phosphodiesterase-5 inhibition improved drug-assisted erectile function and *significantly protected penile length* (2.2 mm loss on daily tadalafil versus 7.9 mm on demand and 6.3 mm placebo), yet **unassisted erectile function after drug-free washout was not improved in any of them**. A 2020 commentary in a leading review journal described the field as “a long list of negative penile rehabilitation trials”. The structural endpoint moved; the functional one did not.

The internal contradiction. The conjecture cites a single case report of mild corporal fibrosis on Doppler. The largest reported series in the same systematic review — 88 men — found **no patient with elevated echogenicity on B-mode imaging**. That is the strongest available evidence on the question and it points the other way.

Tests. Ultrasound elastography or magnetic resonance stratified by symptom duration against age-matched controls remains the right study, and is feasible now. We add two requirements the conjecture omits: a normative dataset for penile tissue stiffness in young men, and formal nocturnal penile tumescence monitoring to establish whether nocturnal erections are genuinely absent rather than merely unnoticed.

Keywords: hard flaccid syndrome; cavernosal hypoxia; transforming growth factor $\beta 1$; corporal fibrosis; nocturnal erections; penile rehabilitation; elastography.

Reader advisory

This is hypothesis generation, not a report of new data. No component of the mechanism proposed here has been demonstrated in a person with hard flaccid syndrome. Nothing in this paper supports taking phosphodiesterase-5 inhibitors, or any other drug, to prevent a fibrotic process that has not been shown to occur in this condition. Nothing here should influence clinical management.

1. Introduction

Most conjectures about hard flaccid syndrome (HFS) try to explain what the condition *is*. This one tries to explain why it does not stop.

The chronicity is the striking feature. Symptom resolution was documented in only 2 of the 10 patients whose outcomes were reported in the 2025 systematic review [B] [1], in a population whose mean age was 27.4 years [B] [2]. Loss of morning erections was reported by 66.2% of surveyed patients [B] [1, 2], and reduced or absent nocturnal erections are described across the case literature [5].

The conjecture assembles these into a loop [5]. Sustained contraction reduces cavernosal perfusion. Loss of nocturnal erections removes the tissue’s principal reoxygenation mechanism. Chronic hypoxia raises transforming growth factor $\beta 1$ (TGF- $\beta 1$). Collagen accumulates. Compliance falls, veno-occlusive function degrades, erections worsen, and nocturnal erections fall further. Each turn makes the next easier and the reverse harder — a ratchet, converting a functional disorder into a structural one.

The logic is attractive and the underlying biology is real. What we argue here is that the conjecture has not noticed that its central claim has already been subjected to a large, expensive and repeated human test in another condition, and that the result of that test is instructive and unfavourable.

2. Methods

This is a hypothesis and theory article. No new human or animal data were generated; no patients were involved; ethical approval was not required.

Literature was identified by targeted searching of MEDLINE/PubMed and the open web between May and August 2026, using combinations of the terms *cavernosal hypoxia fibrosis*, *TGF-beta1 corpus cavernosum collagen*, *cavernous neurotomy fibrosis*, *penile rehabilitation randomised trial*, *nocturnal penile tumescence reoxygenation*, and *hard flaccid syndrome*, with backward citation chasing. The search was purposive and non-systematic. We searched specifically for the clinical trial record of interventions intended to prevent cavernosal hypoxia, and Section 5 reports what that returned.

Claims are graded as in Table 1. Claims not verified against a retrievable source are marked [VERIFY]; quantities that do not exist in the literature are marked [DATA NEEDED].

Table 1. Evidence grading scheme. The grade refers to the strength of evidence for the *stated proposition*, not to its relevance to hard flaccid syndrome. No proposition here carries grade A evidence *in HFS*, because none exists.

Grade	Definition
[A]	Replicated human evidence: randomised trials or consistent findings across independent human studies.
[B]	Single human study, human tissue study, congress abstract, or human data with substantial methodological limitation.
[C]	Experimental animal or in vitro evidence, with species and preparation stated.
[D]	Mechanistic inference: deduced from A–C evidence but not itself directly observed.
[E]	Unverified anecdotal or community-sourced report; not peer reviewed, denominators unknown.

3. The pathway is real

The molecular chain the conjecture invokes is well characterised, and we state it fully before criticising its application.

In the rat, cavernous neurotomy produces significantly greater TGF- $\beta 1$ protein and messenger RNA expression, hypoxia-inducible factor 1α expression, and collagen I and III protein expression than sham surgery [C] [9]. Cavernous nerve ablation produces corporal fibrosis in animals [C] [VERIFY] [10]. Bilateral cavernous neurotomy induces penile hypo-oxygenation demonstrable by hypoxyprobe immunohistochemistry, together with overexpression of the pro-fibrotic endothelin-1 type B receptor [C] [VERIFY] [16, 15]. Penile denervation also produces apoptosis in the rat penis [C] [VERIFY] [14].

The oxygen-dependence is equally well specified. TGF- $\beta 1$ increases collagen synthesis in cultured human corpus cavernosal smooth muscle cells, and a single application of prostaglandin E1 was sufficient to significantly suppress TGF- $\beta 1$ -induced collagen synthesis [C] [VERIFY] [11]. Prostanoid synthesis in this tissue is oxygen-dependent and activates functional prostaglandin E receptors on corpus cavernosum smooth muscle [C] [VERIFY] [12], and cyclic AMP modulates TGF- $\beta 1$ -induced fibrillar collagen synthesis in cultured human corporal smooth muscle [C] [VERIFY] [13]. So the proposed switch — low oxygen favours TGF- $\beta 1$ and collagen, high oxygen favours prostanoids and suppresses collagen — has a real molecular basis [C].

3.1 What the field that developed this theory says about it

The theory is not uncontested in its home discipline, and the conjecture does not report this.

A review of penile rehabilitation states plainly that the data are insufficient to confirm that the balance of TGF- $\beta 1$ and prostaglandin E1 expression actually dictates the varying amounts of smooth muscle and connective tissue within the corpus cavernosum at *physiological* oxygen tensions, and that the role of hypoxia in penile fibrosis after nerve-sparing radical prostatectomy therefore remains a topic of debate. It records the critics’ position directly: the effects on cavernosal tissue of the physiological oxygen tensions associated with a flaccid penis remain theoretical, and **no studies have proven an in vivo derangement of endothelial or smooth muscle cell metabolism secondary to a prolonged**

flaccid state [B] [VERIFY] [17].

That last sentence is decisive for the conjecture as applied to HFS. The animal evidence concerns *surgical denervation*, which produces a degree and duration of hypoxia that a chronically contracted but perfused penis may not approach. The step from “prolonged flaccidity” to “pathological hypoxia” is the weakest link in the chain, and it is unproven in the discipline that proposed it [D].

4. The chain, and what compound probability does to it

The ratchet requires six links in sequence (Figure 1). Each is separately plausible; the conjunction is what the hypothesis asserts, and the evidence for the links is very unequal.

Table 2. The six links of the proposed ratchet, with the evidence available for each in *hard flaccid syndrome*.

#	Link	Evidence	Status
1	Sustained contraction reduces cavernosal perfusion	[D]	Inferred. Penile Doppler peak systolic velocity was normal in all 88 patients in the largest reported series [3].
2	Nocturnal erections are lost	[B]	Self-reported; loss of morning erections in 66.2% [2]. Never confirmed by formal nocturnal penile tumescence monitoring in a cohort.
3	Loss of nocturnal erections produces pathological hypoxia	[D]	The critical link. Contested even in the post-prostatectomy field; no in vivo derangement demonstrated from prolonged flaccidity [17].
4	Hypoxia raises TGF- β 1 and collagen synthesis	[C]	Well supported in animal neurotomy models and human cell culture [9, 11].
5	Collagen accumulation reduces compliance and degrades veno-occlusive function	[C]	Supported in animals and in disease states; not demonstrated in HFS.
6	Reduced compliance further reduces erections, closing the loop	[D]	Entirely inferred.

Only link 4 carries direct experimental support, and it is animal and in vitro. Links 1, 3 and 6 are inferences; link 2 rests on self-report; link 5 has never been examined in this condition. **A chain in which one link is well evidenced and five are not is not a strong hypothesis, however plausible each link seems in isolation [D].**

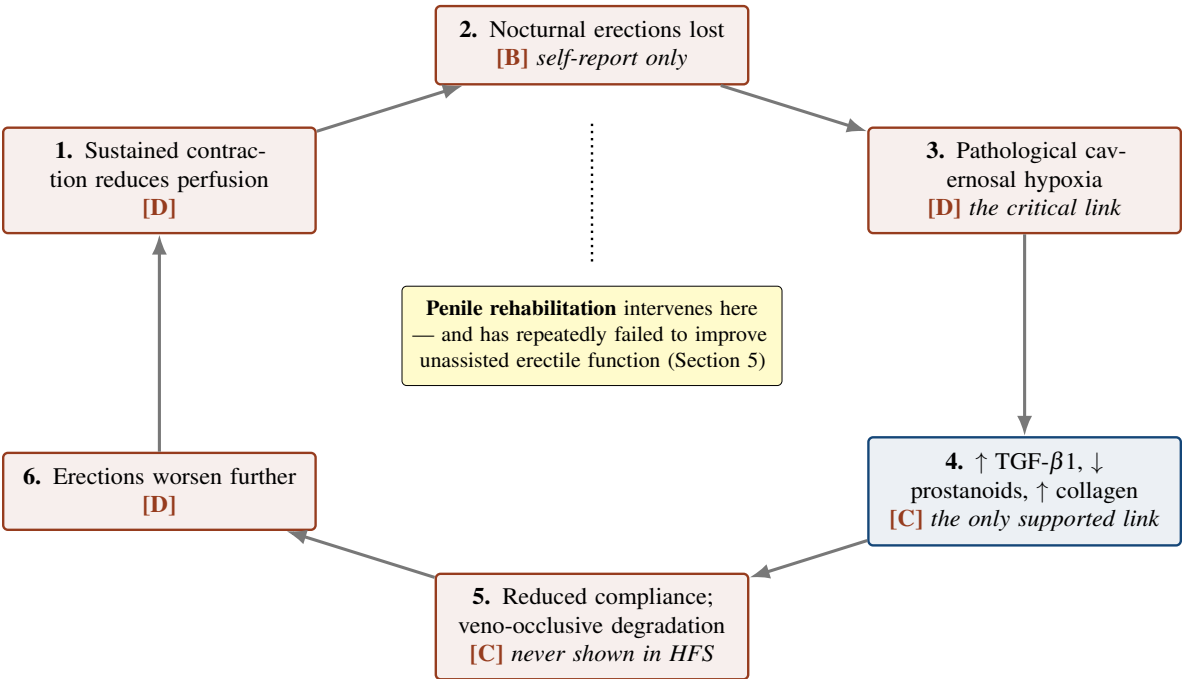


Figure 1. The proposed hypoxia–fibrosis ratchet, with the evidence grade for each link in *hard flaccid syndrome*. Only link 4 — that hypoxia raises transforming growth factor β 1 and collagen synthesis — carries direct experimental support, and that support is from animal neurotomy models and human cell culture. Link 3, the step from prolonged flaccidity to pathological hypoxia, is the one on which everything else depends and is contested in the field that proposed it. The dotted line marks where phosphodiesterase-5 inhibitor rehabilitation intervenes: an intervention designed to restore nocturnal reoxygenation and thereby break the loop.

5. The human test that has already been run

Penile rehabilitation after radical prostatectomy is the hypoxia–fibrosis ratchet turned into a treatment strategy. Its entire rationale is that restoring cavernosal oxygenation during the period of neuropraxia will prevent the fibrotic remodelling that otherwise makes erectile dysfunction permanent [B] [VERIFY] [17]. It has been tested repeatedly, in adequately powered randomised placebo-controlled trials, in a population where the initiating lesion is known and dated.

The results are consistent and instructive [A]:

- **REACTT** randomised 423 men after bilateral nerve-sparing radical prostatectomy to daily tadalafil, on-demand tadalafil or placebo. Daily tadalafil was most effective on drug-assisted erectile function during treatment, and produced significantly less penile length loss (2.2 mm) than on-demand tadalafil (7.9 mm) or placebo (6.3 mm). **Unassisted erectile function after drug-free washout was not improved** [A] [VERIFY: FIGURES TRACED THROUGH A SECONDARY REVIEW; THE PRIMARY REPORT SHOULD BE CONSULTED] [18, 19].
- **REINVENT** found nightly vardenafil efficacious when used on demand but with **no significant effect on unassisted erectile function after drug-free washout** [A] [VERIFY] [20].
- A nightly sildenafil trial was **stopped early because it was expected not to meet its primary endpoint** [A] [VERIFY] [21].
- A 2020 commentary in a leading review journal characterised a further negative trial as the “latest addition to a long list of negative penile rehabilitation trials” [B] [VERIFY] [22].
- A health technology assessment reported that guideline recommendations in this area were often based on evidence that was conflicting, dated, or of low quality [B] [VERIFY] [23].

5.1 What this means for the conjecture

The pattern is not simply negative, and reading it carelessly in either direction would be a mistake. **The structural endpoint moved and the functional endpoint did not.** Daily phosphodiesterase-5 inhibition preserved penile length — a plausible marker of preserved corporal tissue — while failing to improve unassisted erectile function after washout.

Three readings are available [D]:

1. **The ratchet is real but structure is not what determines function.** Fibrosis may accumulate and may be preventable, while the functional deficit is driven by something else — in the prostatectomy setting, the nerve injury itself. Applied to HFS, this would mean the ratchet could be occurring and still not be what maintains the hard flaccid state.
2. **The intervention does not do what it is assumed to do.** Phosphodiesterase-5 inhibition may not restore nocturnal oxygenation sufficiently, in which case the trials do not test the ratchet at all.
3. **The ratchet contributes less than supposed.** If the dominant driver of permanent erectile failure after prostatectomy were preventable hypoxic fibrosis, twenty years of trials directed at exactly that should have produced a clearer signal than they have.

We do not think these can be adjudicated from the existing data, and we note that reading 1 is the most favourable to the conjecture while also being the most damaging to its clinical relevance: *a ratchet that turns but does not determine the symptom is a true mechanism of no explanatory value for HFS.*

6. The internal contradiction

The conjecture’s clinical support is a single published case in which mild corporal fibrosis was found on Doppler — “one patient, but a finding a purely functional model does not anticipate” [5, 4].

The largest reported series in the same systematic review points the other way. Among 88 men with HFS undergoing penile colour Doppler ultrasound, all had normal peak systolic velocity, two (2.3%) had abnormal end-diastolic velocity, and **no patient had elevated echogenicity on B-mode scanning** [B] [3, 1].

Elevated B-mode echogenicity is the conventional sonographic signature of corporal fibrosis. Its complete absence in 88 patients is far stronger evidence than its presence in one, and it is drawn from the same review the conjecture relies on for the case report. **On the currently available imaging evidence, corporal fibrosis in HFS is not merely undemonstrated but positively unsupported** [D].

Two qualifications, both real. B-mode echogenicity is an insensitive measure and would miss early or diffuse fibrosis; the Sullivan data are a congress abstract with no reported control group and no statement of symptom duration. So the correct conclusion is not that fibrosis is absent but that *the one imaging modality applied at scale in this population has*

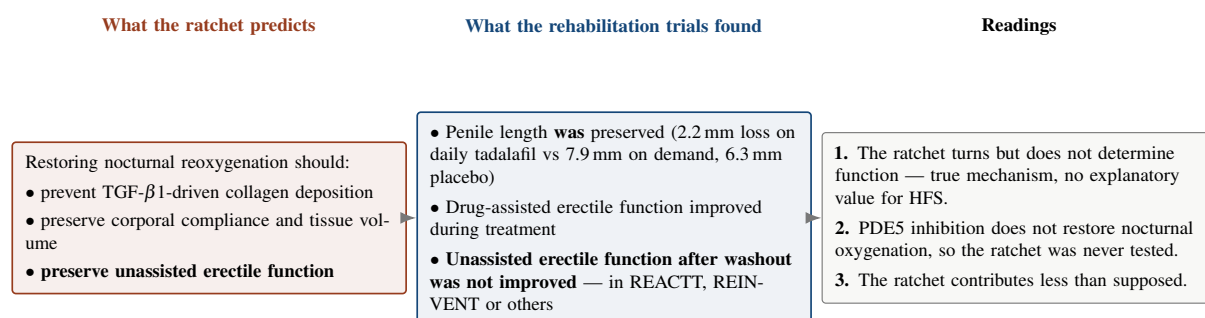


Figure 2. The natural experiment the conjecture does not cite. Penile rehabilitation after radical prostatectomy is an attempt to break exactly this ratchet, in a population where the initiating lesion is known and dated, tested in adequately powered randomised placebo-controlled trials. The structural endpoint responded; the functional endpoint did not. Note that the reading most favourable to the conjecture’s biology — that fibrosis accumulates but is not what drives the symptom — is the reading least favourable to its relevance.

not detected it, and that a more sensitive modality is required before the conjecture’s central structural claim can be evaluated.

7. The hypothesis

Formal statement

H10 (hypoxia–fibrosis ratchet). In hard flaccid syndrome, loss of nocturnal erections removes the tissue’s principal reoxygenation mechanism; the resulting chronic cavernosal hypoxia drives TGF- β 1-mediated collagen deposition; the resulting loss of compliance further impairs erection and reoxygenation, so that the disorder progresses from functional to structural and does so asymmetrically.

H10-strong. The ratchet is what maintains the hard flaccid state in established disease. Prediction: measurable corporal fibrosis proportional to symptom duration; treatment response declining with duration; the state irreversible once fibrosis is established.

H10-weak (the form the rehabilitation trials leave open). The ratchet turns — fibrosis accumulates with time — but is not what maintains the sign. Prediction: fibrosis detectable and duration-dependent, but not correlated with the severity of the hard flaccid state, and preventable without abolishing the symptom.

H10-null. No excess corporal fibrosis occurs. Chronicity has another explanation.

Auxiliary assumptions:

- **A1 (nocturnal erections are genuinely lost).** Reported by self-report only; never established by formal nocturnal penile tumescence monitoring in an HFS cohort [DATA NEEDED]. This is cheaply checkable and the whole ratchet starts here.
- **A2 (flaccidity in HFS is more hypoxic than ordinary flaccidity).** Cavernosal oxygen tension has never been measured in this condition [DATA NEEDED]. Note that the Doppler data show *normal* arterial inflow [3], which does not obviously support reduced perfusion.
- **A3 (the timescale fits).** Fibrotic remodelling sufficient to change organ mechanics must occur within the disease duration of a young population. Untested.

8. Predictions and tests

8.1 The prerequisite the conjecture omits

Prediction P1. Formal nocturnal penile tumescence and rigidity monitoring will confirm reduced or absent sleep-related erections in men with HFS relative to age-matched norms.

The entire ratchet begins with a premise established only by self-report. Nocturnal penile tumescence monitoring is an established technique, has been applied to two HFS patients in the published literature with suboptimal erections reported in one [B] [1], and is non-invasive. **If nocturnal erections are preserved, the ratchet does not start** — and men who monitor their symptoms closely may notice the absence of *morning* erections without that reflecting absence of nocturnal ones.

8.2 The primary structural test

Prediction P2. Corporal tissue stiffness, measured by shear-wave ultrasound elastography or magnetic resonance, is increased in men with HFS relative to age-matched controls, and the increase correlates with symptom duration.

This is the conjecture’s own proposed test [5] and it is the right one. Three additions are required to make it interpretable:

- **Age-matched controls and normative data.** Corporal stiffness varies with age; we identified no normative dataset for penile elastography in young men [DATA NEEDED].
- **Duration stratification designed in from the start,** since the hypothesis is fundamentally about a time course and a cross-sectional comparison without duration data cannot test it.
- **Standardised tumescence state at measurement,** since tissue stiffness in a distensible organ depends on filling.

Prediction P3. Response to phosphodiesterase-5 inhibitors and to α -blockers declines with symptom duration.

Checkable retrospectively from existing records. We note, as in two companion papers, that this prediction is shared with other conjectures and would support a class rather than this hypothesis specifically [7, 8].

8.3 The measurement nobody has made

Prediction P4. Cavernosal oxygen tension in the resting hard flaccid penis is lower than in age-matched controls.

Intracavernosal blood gas sampling has been performed historically in the erectile physiology literature, and is the only direct test of link 3. It is invasive, requires needle puncture of the corpus cavernosum, and in a young population with a non-life-threatening condition would need careful justification. Transcutaneous or near-infrared spectroscopic approaches would be preferable if they can be validated for this tissue [DATA NEEDED]. We note it because **the hypothesis is named after a variable nobody has measured in the patients it concerns.**

Table 3. Predictions of H10 compared with the principal alternatives.

Observation	H10 (ratchet)	Effector autonomy (tone)	Neural drive	Discriminating value
Increased corporal stiffness on elastography	Predicted	Not predicted	Not predicted	High
Stiffness correlates with symptom duration	Predicted	Not predicted	Not predicted	High
Elevated B-mode echogenicity	Predicted	Not predicted	Not predicted	High — and absent in 88/88 [3]
Reduced cavernosal oxygen tension at rest	Predicted	Not predicted	Not predicted	High, and never measured
Absent nocturnal erections on formal monitoring	Predicted (required)	Predicted	Predicted	None
Declining drug response with duration	Predicted	Predicted	Not predicted	Low
Normal arterial inflow on Doppler	Not predicted (link 1)	Predicted	Predicted	Moderate, against H10

9. Alternative explanations

1. **Chronicity without structural change.** Every other conjecture in this series proposes a maintaining mechanism that is functional rather than structural — a sensitised spinal circuit, a calcium-sensitised effector, an ectopic generator [6, 7]. None requires fibrosis, and none is excluded by the low remission rate.
2. **The low remission rate may be an artefact of ascertainment.** Men whose symptoms resolve leave patient communities and stop seeking care. A resolution rate calculated from case reports and self-selected surveys will systematically underestimate remission, and the “20%” figure derives from 2 of 10 patients in published case material [1]. **This is a simpler explanation of apparent irreversibility than a ratchet.**
3. **Reverse causation.** If fibrosis is eventually found, it may be a consequence of the sustained contraction rather than a cause of its persistence — a marker of duration rather than a mechanism.
4. **Psychological and behavioural maintenance.** Depression or anxiety was reported in 75% of the largest series [B] [3], and hypervigilant monitoring of a genital sign is itself a maintaining process requiring no tissue change.
5. **The premise may be false.** Cavernosal tone has never been objectively shown to be elevated in HFS, and cavernosal oxygenation has never been measured in it.

10. Implications

No clinical implications are drawn. In particular, nothing here supports prophylactic phosphodiesterase-5 inhibitor use to prevent a fibrotic process that has not been shown to occur in this condition, and the trial record in the setting where that strategy has actually been tested is discouraging.

Three research implications follow.

First, check that the ratchet starts. Formal nocturnal penile tumescence monitoring in an HFS cohort is non-invasive, uses established technology, and tests the premise on which everything else depends.

Second, read the rehabilitation literature. Any hypothesis proposing that cavernosal hypoxia drives an irreversible fibrotic decline should engage with twenty years of randomised trials testing exactly that proposition in another condition.

Third, use a sensitive modality. B-mode echogenicity has been applied at scale and found nothing; elastography or magnetic resonance is required, with age-matched controls and duration stratification built in.

11. Limitations

1. **No evidence in HFS.** Cavernosal oxygen tension, TGF- β 1 expression and corporal collagen content have never been measured in this condition.
2. **Five of six links are unsupported** in this population (Table 2), and the conjunction is what the hypothesis asserts.
3. **The critical link is contested in its home field.** The effects of physiological flaccid-state oxygen tensions on cavernosal tissue remain theoretical [17].
4. **The available imaging evidence points the other way:** no elevated echogenicity in 88 patients, against one case report.
5. **Normal arterial inflow** in the same 88 patients does not support the reduced-perfusion premise of link 1.
6. **The animal evidence is from surgical denervation,** a lesion of different kind and severity from anything proposed in HFS.
7. **The low remission rate has a simpler explanation** in ascertainment bias.
8. **The trial figures are second-hand.** The REACTT length-loss numbers were traced through a review and should be confirmed against the primary publication.
9. **Several citations are flagged unverified,** particularly the older cell-culture work and the rehabilitation trial reports.
10. **Confirmation risk.** This paper was written to develop a specific conjecture and concludes largely against it. Readers should check the rehabilitation trial record for themselves rather than accepting our characterisation of it, since that characterisation carries most of the argument.

12. Refutation criteria

- **Refuted.** Elastography or magnetic resonance shows no increase in corporal stiffness in men with HFS relative to age-matched controls, at any symptom duration.
- **Substantially refuted.** Formal nocturnal penile tumescence monitoring shows preserved sleep-related erections in men with HFS.
- **Substantially refuted.** Cavernosal oxygen tension in the resting hard flaccid penis is normal.
- **Weakened.** Corporal stiffness, if increased, shows no relationship to symptom duration — indicating a marker rather than a progressive process.
- **Demoted to H10-weak.** Fibrosis is demonstrated and duration-dependent, but severity of the hard flaccid state is uncorrelated with it.
- **Rendered moot.** Direct measurement shows resting cavernosal tone is not objectively elevated in symptomatic patients.

13. Conclusion

This conjecture proposes a mechanism whose molecular components are among the best characterised in erectile biology. Cavernous neurotomy in the rat does raise hypoxia-inducible factor 1 α , transforming growth factor β 1 and collagen; oxygen-dependent prostanoid synthesis does suppress TGF- β 1-driven collagen deposition. The chemistry is not in doubt.

What is in doubt is everything else. The ratchet needs six links and only one of them has direct experimental support. Its critical step — that prolonged flaccidity produces pathological hypoxia — is contested in the very field that developed the theory, which records that no study has demonstrated in vivo derangement of cavernosal metabolism from

a prolonged flaccid state. The premise that nocturnal erections are lost rests on self-report and has never been confirmed by monitoring. And the sole clinical support offered, a single case of mild fibrosis on Doppler, sits alongside a series of 88 men in which no patient had elevated echogenicity at all.

The most important omission is larger than any of these. The hypoxia–fibrosis ratchet has already been converted into a treatment strategy and tested repeatedly in adequately powered randomised trials — that is what penile rehabilitation after radical prostatectomy is. Those trials preserved penile length and improved drug-assisted erections, and failed, consistently, to improve unassisted erectile function after washout. A commentary in a leading review journal describes the field as a long list of negative trials. The structural endpoint moved and the functional one did not, and the reading of that result most favourable to the conjecture’s biology — that fibrosis accumulates but is not what drives the symptom — is precisely the reading that makes it irrelevant to hard flaccid syndrome.

Two studies would settle most of this and neither is difficult. Put a cohort on nocturnal tumescence monitoring and find out whether the ratchet starts. Then image them with elastography, stratified by duration, against age-matched controls, and find out whether it turns. Until then, the field is entitled to note that the condition rarely remits, and not entitled to conclude that it is because the tissue has changed.

Declarations

Authorship and the use of artificial intelligence. The second listed author is a large language model. It is listed at the corresponding author’s request and in the interest of transparency, but this contravenes the recommendations of the International Committee of Medical Journal Editors, which hold that artificial intelligence tools cannot meet authorship criteria because they cannot take responsibility for the accuracy and integrity of the work. **Before submission to any peer-reviewed venue this attribution should be converted into an AI-use disclosure, with the human author accepting sole responsibility for the content.** The penile rehabilitation trial results carry most of the argument in this paper and were traced through reviews and commentaries; the primary trial reports must be read and the figures confirmed before submission.

Ethics approval. Not applicable. No human participants, animals or identifiable data were involved.

Funding. None.

Competing interests. No competing financial interests. The first author maintains a public website on hard flaccid syndrome and authored the source article [5]; this is a non-financial interest in the hypothesis being taken seriously, and readers should weigh the argument accordingly.

Data availability. No datasets were generated or analysed.

Clinical disclaimer. This article is hypothesis generation. It does not constitute medical advice and does not establish a clinician–patient relationship. **Nothing in it supports the use of phosphodiesterase-5 inhibitors or any other agent to prevent corporal fibrosis in hard flaccid syndrome**, a process that has not been demonstrated to occur in this condition. Intracavernosal blood sampling and pharmacological erection testing are invasive procedures appropriate only under qualified medical supervision.

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